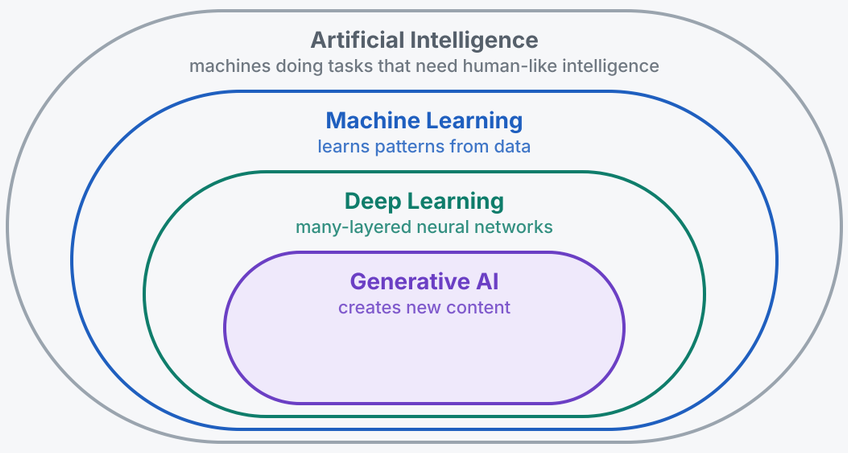
Machine Learning Algorithms: A Visual Framework

12 algorithms organized by how they learn, where they are used, and how they relate to each other

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LEARNING STYLE Supervised · labeled data | Unsupervised · no labels | Unsupervised · self-supervised generative

DOMAIN Tabular | Computer Vision | NLP | Generative AI

Supervised Learning

Learns from examples that already have the right answer, then predicts the answer for new data.

CLASSIC MACHINE LEARNING · STRONGEST ON TABULAR DATA (ROWS AND COLUMNS)

1 Linear Regression

predicts a number

Supervised · Tabular

HOW IT WORKS Fits the straight line (or plane) that keeps the gap between predicted and actual values as small as possible.

- Forecasting monthly sales
- Estimating house prices from size and location

2 Decision Tree

classifies or predicts

Supervised · Tabular

HOW IT WORKS Asks a series of yes/no questions about the features, choosing each split to make the groups as pure as possible.

- Loan approval rules a bank can explain
- Sorting fruit by shape, color, and texture

3 Random Forest

ensemble of trees

Supervised · Tabular

HOW IT WORKS Trains hundreds of trees on random samples of rows and features, then lets them vote. Averaging cuts overfitting.

- Credit card fraud detection
- Predicting which customers will cancel

4 Gradient Boosting

XGBoost, LightGBM

Supervised · Tabular

HOW IT WORKS Builds small trees one after another; each new tree focuses on correcting the errors the earlier trees made.

- Credit risk scoring
- Demand forecasting for retail inventory

5 Support Vector Machine

SVM

Supervised · Tabular · NLP

HOW IT WORKS Finds the boundary that separates two classes with the widest possible margin between them.

- Classifying emails or news articles by topic
- Detecting faulty parts from sensor readings

6 K-Nearest Neighbors

KNN

Supervised · Tabular · Vision

HOW IT WORKS Labels a new point by checking the K most similar known examples and taking the majority answer.

- "Customers like you also bought" suggestions
- Finding visually similar product images

How 2, 3 and 4 relate: one **Decision Tree** → many trees voting in parallel = **Random Forest** → trees built in sequence to fix errors = **Gradient Boosting**. More trees usually mean more accuracy but less explainability.

DEEP LEARNING · LEARNS ITS OWN FEATURES FROM RAW PIXELS

7 Convolutional Neural Network

CNN

Supervised · Computer Vision · Generative AI

HOW IT WORKS Slides small filters across an image; early layers detect edges, later layers combine them into shapes and whole objects. Unlike classic ML, nobody has to hand-design the features.

- Spotting tumors in medical scans
- Detecting pedestrians for self-driving cars
- Serves as the image backbone inside many GANs and diffusion models

CLASSIC ML VS. DEEP LEARNING

Classic ML needs people to define features (size, color, price). It works well on smaller, structured datasets and is easier to explain.

Deep learning discovers features itself, so it excels at images, audio, and text, but needs much more data and computing power.

Unsupervised Learning

Finds structure in data that has no labels.

8 K-Means Clustering

groups similar items

Unsupervised · Tabular · Vision

HOW IT WORKS Places K centers, assigns every point to its nearest center, moves each center to the middle of its group, and repeats until stable.

- Customer segmentation for marketing
- Reducing an image to a few colors

9 Principal Component Analysis

PCA

Unsupervised · Tabular · Vision

HOW IT WORKS Finds the few directions in which the data varies most and re-describes each item using only those, shrinking many columns into a handful.

- Compressing hundreds of features before modeling
- Plotting high-dimensional data in 2-D

Generative AI · language

Learns from unlabeled text, then creates new text.

10 Transformer

GPT, BERT, Claude

Self-supervised, then fine-tuned · NLP · Generative AI · Vision

HOW IT WORKS Splits text into tokens and uses *attention* to weigh how every token relates to every other. Pretrained by predicting missing or next words, so raw text supplies its own labels.

- Chatbots, translation, summarizing documents
- Sentiment analysis of customer reviews

Generative AI · images

Two different ways to teach a network to create realistic images.

11 Generative Adversarial Network

GAN

Unsupervised · Generative AI · Computer Vision

HOW IT WORKS Two networks compete: a *generator* makes fake images and a *discriminator* tries to catch them. Each improves until the fakes look real.

- Upscaling low-resolution photos
- Creating synthetic training images when real data is scarce

12 Diffusion Model

Stable Diffusion, DALL·E

Unsupervised · Generative AI · Computer Vision

HOW IT WORKS Learns to remove noise step by step. To create, it starts from pure static and denoises it into an image, guided by a text prompt.

- Text-to-image design and marketing art
- Filling in or editing parts of a photo

Where each algorithm is used

Filled dot = primary domain · ring = also used there

ALGORITHM	TABULAR	VISION	NLP	GEN AI
1 Linear Regression	●			
2 Decision Tree	●			
3 Random Forest	●			
4 Gradient Boosting	●			
5 SVM	●	○	○	
6 K-Nearest Neighbors	●	○		
7 CNN		●		○
8 K-Means	●	○		
9 PCA	●	○		
10 Transformer		○	●	●
11 GAN		●		●
12 Diffusion Model		●		●

Which one should I try first?

A quick starting point, not a strict rule

1. Do you want to create new content?

YES — **Text:** Transformer (10) · **Images:** Diffusion (12) or GAN (11)

2. No. Does your data have labeled answers?

NO LABELS — **Find groups:** K-Means (8) · **Simplify many columns:** PCA (9)

LABELED — **What kind of data?**
Rows and columns → start with Gradient Boosting (4) or Random Forest (3); use a Decision Tree (2) when you must explain every decision, Linear Regression (1) for a simple number forecast
Images → CNN (7) · Text → fine-tuned Transformer (10)

Key sources: Breiman (2001), Random forests · Chen & Guestrin (2016), XGBoost · Cortes & Vapnik (1995), Support-vector networks · LeCun et al. (1998), Gradient-based learning applied to document recognition · Goodfellow et al. (2014), Generative adversarial nets · Vaswani et al. (2017), Attention is all you need · Ho et al. (2020), Denoising diffusion probabilistic models · 3Blue1Brown (2024) · StatQuest with Josh Starmer (2021). Full references are listed in the artifact description.